DIGITAL LIFESTYLE AND PRODUCTIVITY

An Exploratory Data Analysis of Social Media Usage and Productivity

Kanishka Gowda N 1RUA25SCS0037 | Yuktha H 1RUA25SCS0120 | Niharikaa Dhanya Kumar 1RUA25SCS0064 | Navya TS 1RUA25SCS0062 | Afreen Unnisa 1RUA25SCS0004

Introduction

This study explores how social media usage relates to individual productivity by analysing digital habits and productivity patterns.

Objectives

- Analyse social media usage patterns.
- Examine its relationship with productivity.
- Identify trends through data visualization.

Dataset

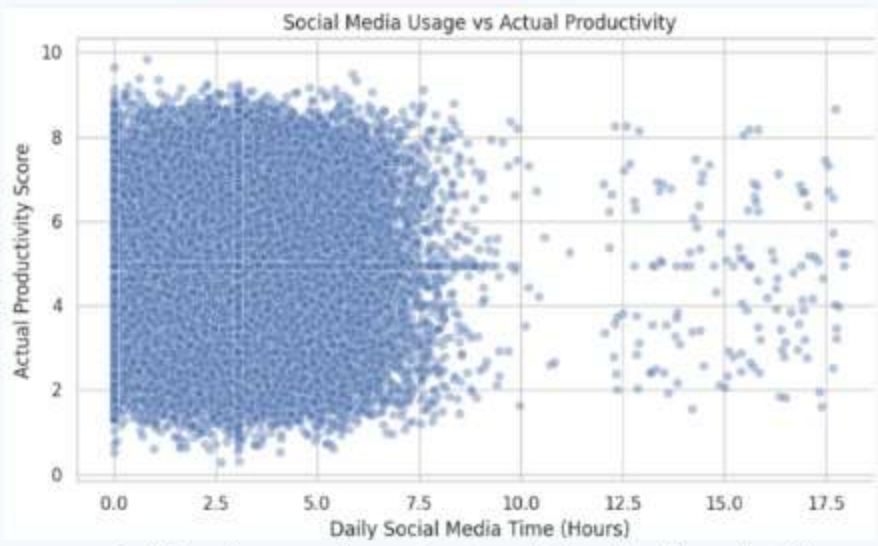
- Records: 30,000 entries
- Features: 19 columns
- Content: Includes social media usage habits, screen time and productivity-related information.
- Dataset Source: https://www.kaggle.com/datasets/mahdimashayekhi/social-media-vs-productivity

Methodology

- Data Collection: Obtained a dataset containing information about social media usage and productivity.
- Data Preprocessing: Checked for missing values and duplicate records and cleaned the dataset.
- Data Analysis: Used Python, Pandas and NumPy to analyse usage patterns and productivity metrics.
- Data Visualization: Created graphs using Matplotlib and Seaborn to identify trends and patterns.
- Correlation Analysis: Examined the relationship between daily social media usage and actual productivity.

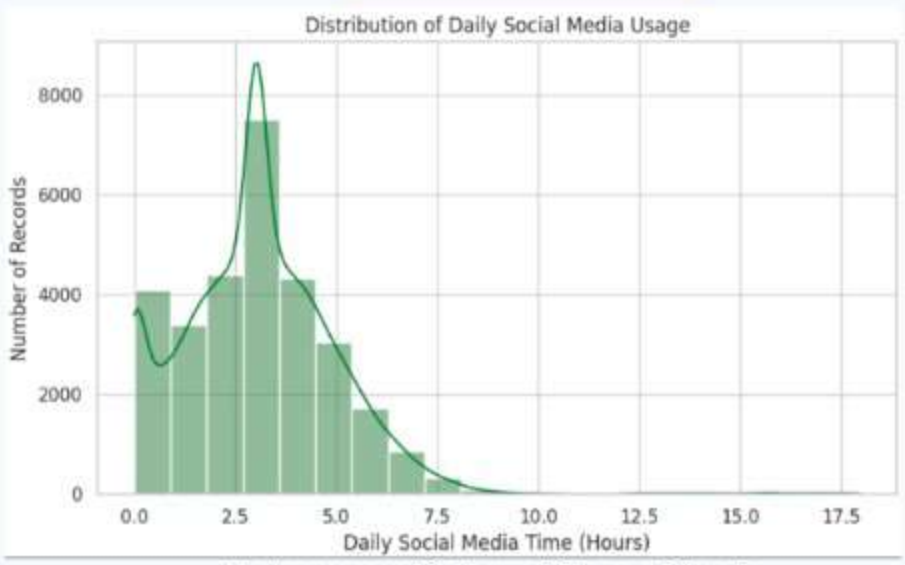
Visual Analysis

Scatter Plot



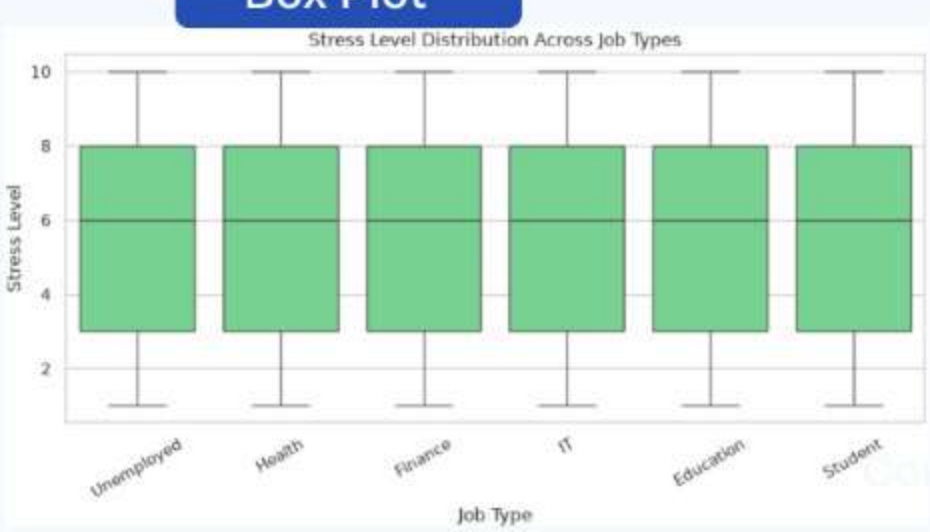
Social media usage shows almost no relationship with productivity.

Histogram



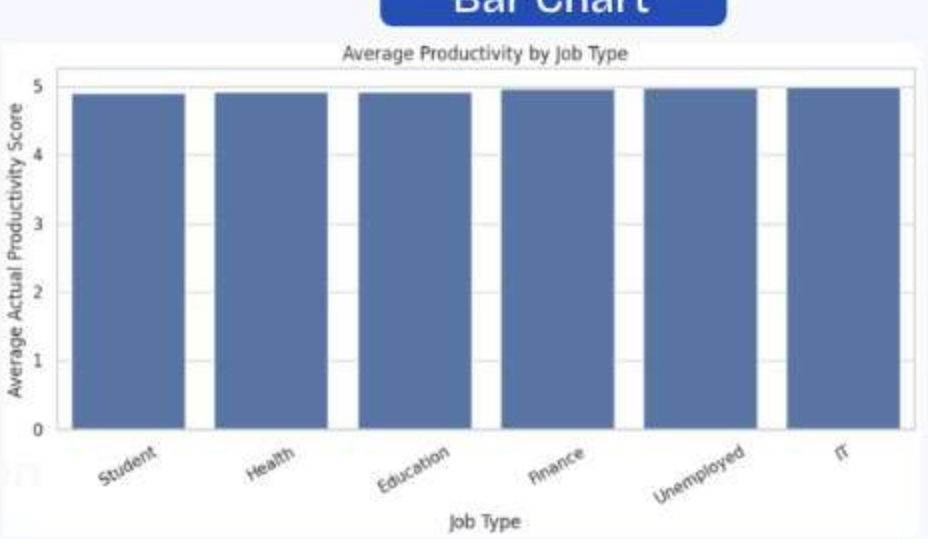
Most users spend 2–4 hours daily on social media.

Box Plot



Stress levels remain similar across all job types.

Bar Chart



Average productivity is nearly equal across all job types.

Key Findings

- Dataset contained 30,000 records.
- 16,187 missing values were identified.
- Social media usage had a negligible negative correlation with productivity (-0.0101).

Recommendations

- Reduce unnecessary screen time.
- Practise effective time management.
- Maintain balanced digital habits.

Conclusions

Social media usage alone does not determine productivity. Other lifestyle factors also play an important role in influencing productivity.